

Historical partial packet: the ℓ_2 nilpotent right-perturbation subcase

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Status

This is a historical partial packet. It was absorbed into the full packet

`solutions/full/1605.07409_nilpotent_right_perturbation_derivations/`.

Source question

The source paper asks in Section 6 whether reasonable sufficient conditions exist on a pair (A, B) inducing hypercyclic generalized derivations $\tau_{A,B} = L_A - R_B$ on separable Banach ideals.

6. FURTHER QUESTIONS

Some natural questions arising from this paper include the following.

1. Do reasonable sufficient conditions exist on the pair (A, B) that induce hypercyclic $\tau_{A,B}$ on separable Banach ideals?
2. Corollary 5.2 and Example 5.3 contrast the hypercyclic properties of

Historical partial theorem

Let $X = \ell_2$, let W_λ be λ times the backward shift for some $\lambda > 1$, and let $N \in \mathcal{L}(X)$ be nonzero and nilpotent. Then, for every separable admissible Banach ideal $J \subset \mathcal{L}(X)$, the generalized derivation

$$\tau_{W_\lambda, N} = L_{W_\lambda} - R_N$$

satisfies the Hypercyclicity Criterion.

Promotion

The full packet proves a stronger abstract criterion: W_λ may be replaced by any operator A with a right inverse F , $AF = I$, $r(F) < 1$, and a dense subspace eventually killed by powers of A . This historical subcase is retained only to document the partial-to-full progression.